ARE THE PERFECT SQUARES A STATIONARY SET?

ØISTEIN SØVIK

Evidence is labelled Sketch: this proof has not been posted on arXiv or peer reviewed and should be checked by the reader. AI assistance was used in adapting and checking the argument and exposition.

Is the set S of perfect squares stationary?

Source: Hervé Queffélec, "Dirichlet Series and Function Theory in Polydiscs," Oberwolfach Report 06/2014, doi:10.4171/OWR/2014/06, Problem Session, §3.1, Question 1, pp. 385–386 (reported by O. F. Brevig).

For a trigonometric polynomial $P(t) = \sum_k a_k e^{ikt}$, write $[P] = \mathbb{E}\|\sum_k \varepsilon_k a_k e^{ikt}\|_\infty$, where (ε_k) are independent Rademacher variables. A set $E \subset \mathbb{Z}$ is stationary if $[P] \le C\|P\|_\infty$ for all trigonometric polynomials supported in E, with C independent of P.

Answered: no. Evidence: Sketch.

No; even the odd positive squares are not stationary.

Proof. Choose $w \in C_c^\infty((1/2,1))$, with $0 \le w \le 1$ and $w = 1$ on $[3/5, 4/5]$, and put

$$P_N(t) = \sum_{n\ge 1} \sin(\pi n/2) w(n/N) e^{in^2 t}.$$

The theta identity and modular formulas (NIST *Digital Library of Mathematical Functions*, https://dlmf.nist.gov/23.18) give the identity $\eta(\tau)^3 = \sum_{j\ge 0}(-1)^j(2j+1)e^{\pi i\tau(2j+1)^2/4}$ and the modular transformations of η imply

$$\left|\sum_{n\ge 1} n \sin(\pi n/2) e^{-yn^2} e^{in^2 t}\right| \le C y^{-3/4}$$

uniformly in t. Put

$$F_y(t) = \sum_{n\ge 1} \sin(\pi n/2) e^{-yn^2} e^{in^2 t}.$$

Integrating the weighted estimate against $v^{-1/2}$ gives $\|F_y\|_\infty \le C y^{-1/4}$. Set $m(u) = e^u w(\sqrt{u})$ on the support of $w(\sqrt{u})$ and zero elsewhere. The smooth cutoff $K_R(t) = \sum_{k\in\mathbb{Z}} m(k/R) e^{ikt}$ satisfies

$$|K_R(t)| \le C_w \min\{R, 1/(Rt^2)\}, \qquad -\pi < t \le \pi,$$

by taking two differences of its coefficients. Thus $\|K_R\|_1$ is uniformly bounded. The exact multiplier identity $P_N = F_{N^{-2}} * K_{N^2}$ gives $\|P_N\|_\infty \le C_w\sqrt{N}$.

Lefèvre, *On some properties of the class of stationary sets*, doi:10.4064/cm-76-1-1-18, Lemma 2.1, p. 5, gives $[P] \ge c\delta\sqrt{M \log M}$ when at least M Fourier coefficients of P have modulus at least δ. There are at least $N/20$ odd $n \in [3N/5, 4N/5]$ whose coefficients in P_N have modulus one. Thus $[P_N] \ge c_w\sqrt{N \log N}$ and $[P_N]/\|P_N\|_\infty \to \infty$. Every frequency is an odd square, proving nonstationarity. □

Date: 13 October 2026.